

TrES-2b

R-2212

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_111008 · created 2026-08-14T22:52:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455842.71333 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.015
Transit depth (Rp/Rs)^2	2.22 +/- 0.44 [%]
Orbital Inclination (inc)	83.22 +/- 0.3
Airmass coefficient 1 (a1)	1.0064 +/- 0.0097
Airmass coefficient 2 (a2)	-0.0041 +/- 0.004
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	#3 - [290, 470]
Optimal Aperture	3.83
Optimal Annulus	11.85
Transit Duration (day)	0.0668 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.015 vs 0.1254 (deviation 1.48σ)
Duration fit vs published	0.0668 d vs 0.0746 d (deviation 1.15σ)
Mid-time O–C	-0.3 min
Depth SNR	9.4
Residual scatter	0.96 %

Light curve

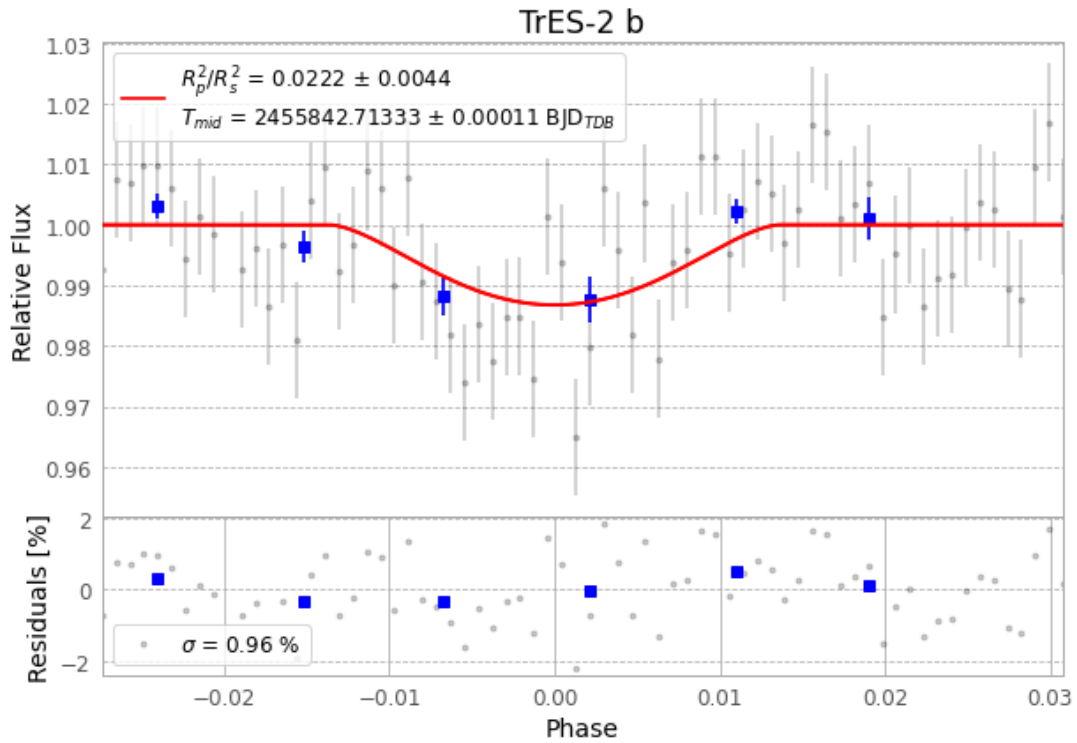


Figure 1 — FinalLightCurve_TrES-2 b_2011-10-07.png (SHA-256 0a34a6515501fbeb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [229, 312], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ef0796df2d3e8e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), TRES-2111008032712.FITS → TRES-2111008065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-111008.log (5dd38f2b72c9...), FinalLightCurve_TrES-2 b_2011-10-07.png (0a34a6515501...), FinalLightCurve_TrES-2 b_2011-10-07.csv (e3fedbd70ee7...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 22:52Z.